--Question:

How can I compute statistics from drillhole assays?

--Answer:

There are two challenges when doing statistical analysis of drillholes assays.

1- The samples are not the same length; so it has different support in geostatistics terms. We have two options; the first one is to weight each sample by its length. The second one is to composite the assays into regular interval.
2- The drillholes are spatially clustered, that it preferentially located in typically zones of high interests. This creates a statistical bias that often overestimate the mean and the variance of the data. The solution is to first decluster the data with a geostatistical approach. Note that delustering is typically done on compositing data.

--Question:

What is the challenge with resource estimation of gold deposit

--Answer:

Gold is particularly difficult since the value of a deposit is often controlled by the a few percentages of the samples that shows very high gold grade content. This create very positively skewed distribution (lots of low value and few high values) and variograms that are difficult to model. Changing the a few high-grade samples such as with capping for instance can have a significant impact on the resource.

--Question:

Why is domaining for resource estimation?

--Answer

Domaining is the process of partitioning a deposit into several parts that are statistically amenable for geostatistics. Resource estimation is not the only reason to do domaining, it can also be done for geological and geometallurgical studies. An often-used approach is to do the spatial portioning using the geology of the deposit, that is segmenting the deposit space by geology. Leapfrog is often used in that context. The assumption here, is that the grade within a geological domain is well-behaved and can be used for geostatistical studies.

--Question:

What is a swath plot and how to interpret it?

--Answer:

A swath plot is a graphical tool used in geostatistical studies to validate block models. It helps in comparing the estimated values from the block model against the actual sample data along a specific direction or swath. This comparison is crucial for ensuring that the block model accurately represents the spatial distribution of the variable of interest.

Role of a Swath Plot in Block Model Validation

-Visual Comparison: Swath plots provide a visual comparison between the estimated values from the block model and the actual sample data. This helps in identifying any discrepancies or biases in the model.

-Spatial Trends: They help in understanding the spatial trends and continuity of the variable being modeled. Any significant deviations between the model and the sample data can indicate issues with the model.

-Model Accuracy: By comparing the block model estimates with the actual data, swath plots help in assessing the accuracy and reliability of the block model.

Steps to Interpret a Swath Plot

-Generate the Swath Plot: Create a swath plot by plotting the estimated values from the block model and the actual sample data along a specific direction (e.g., north-south, east-west, or vertical).

-Compare Trends: Look for overall trends in both the block model estimates and the sample data. The trends should generally match if the block model is accurate.

-Identify Deviations: Identify any significant deviations between the block model estimates and the sample data. Large deviations may indicate issues with the model, such as overestimation or underestimation.

-Check for Bias: Assess whether there is any systematic bias in the block model. For example, if the block model consistently overestimates or underestimates the values compared to the sample data, this indicates a bias.

-Evaluate Spatial Continuity: Ensure that the spatial continuity of the variable is well-represented in the block model. The swath plot should show smooth transitions without abrupt changes unless justified by the geology.

-Adjust the Model: If significant issues are identified, adjust the block model parameters or the estimation method to improve the model's accuracy and reduce discrepancies.

By following these steps, you can effectively use swath plots to validate and improve the quality of your block model in geostatistical studies.

--Question:

What is the difference between an estimation and a geostatistical simulation?

--Answer:

Two common geostatistical methods are kriging and conditional simulation. Kriging provides the best linear unbiased local prediction by minimizing the estimation variance, while conditional simulation generates multiple equally probable realizations to assess uncertainty. Kriging should be selected when having local precision is important, but we do not need to characterize the uncertainty, or we do not need to represent the true spatial heterogeneity (patterns or texture) of the phenomenon. In contrast, one should choose simulation whenever the spatial context is important and that uncertainty is important for decision making.

Note that can get local accuracy with a very large set of simulations.

--Question:

Why do we need a variogram model to do a geostatistical estimation and simulation?

--Answer:

Kriging and simulations require variograms that describe the spatial correlation structure of the data. The variogram is a fundamental tool in geostatistics that quantifies how data similarity decreases with distance. It is used to model the spatial continuity and anisotropy and to inform the weighting of sampled data points in the estimation process. By fitting a variogram model to the empirical variogram, geostatisticians can make more accurate predictions and better understand the spatial variability of the studied phenomenon.

--Question:

Does inverse distance weighting needs a variogram mode?

--Answer:

No. Inverse distance weighting (IDW) is a simpler estimation technique that does not use a variogram and is not typically as accurate as Kriging.

--Question:

Describe a typical geostatistical simulation workflow?

--Answer:

Geostatistical workflows typically start with data exploration. domaining and declustering to removes biases in the data sampling to get an accurate mean and variance of the samples. Removing sampling biases are important to estimate the mean grade and the tonnage of the deposit.

The typical estimation continues with variogram modelling. The samples and the data will then be used to parameterize the Kriging function to interpolate the sample on a block model. The validation of an estimation model consists of checking the swath plots and to compare the statistics between the estimated values and the sample values.

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Once declustered, the typical simulation workflow continues with normal score transformation using the desclustering weights. Normal score transformation transform any data distribution into a standard Gaussian (Normal) distribution; the transformed values are called normal scores. A variogram model is then fitted to the normal scores. The normal scores and the variogram is then passed to the simulation algorithms to generate several realizations. These realizations are then back-transformed into grade values. The realizations are validated with swath plot, QQ-plot, and by checking the variogram reproduction. A goal of the realization is to reproduce the declustered distribution and the variogram of the samples.

--Question:

How many geostatistical simulations should I do?

--Answer:

The number of simulations depends on the objective of the study, the density of the data and the spatial continuity of the grades.

For instance, for simple statistical validation, a few, say three to ten, is enough to analyze their quality in terms of the reproduction of the grade distribution and variogram (spatial continuity).

Once, a modeler can create simulation of high quality, 100 simulations is typically a good start given the speed of performing simulation on Seequent Evo. The right number should be viewed in terms of the objectives and the decision-making inherent to the modeling exercise. Would adding more simulation to the ensemble would change the decisions? One can test that by dividing the ensemble of 100 simulations in two ensembles of fifty and make the decision for each ensemble? If each ensemble drives to different decision, then one should seriously consider adding another set of 100 simulations to the ensemble. That analysis can then be repeated with the augmented ensemble.

--Question:

How can I characterize the geological uncertainty?

--Answer:

The two main approaches are

1- Create several geological models (scenarios) with Leapfrog. Each model represents a possible interpretation of the underlying geology.
2- Simulate the geology using a geostatistical approach such as indicator simulation or plurigaussian simulations.

--Question:

What is a normal score transform of the data and when is it necessary?

--Answer:

Normal score transform or Gaussian anamorphosis is the process of transforming any distribution into a Gaussian distribution. There are several ways of achieving that; the most common is to do a quantile-to-quantile transform whereas the each sample is transformed into a Gaussian deviate with the same quantiles. The challenge in this technique is the application of the declustering weights and the duplicate of data. It is common that a sizeable percentage of data will have the same value, e.g. detection limit or zeros, these must then go through a tie-breaking process to be transformed into a Gaussian distribution.

Normal score transform is important because many simulation algorithms, such as turning bands or sequential gaussian simulation, only work in the Gaussian space. The workflow is then to transform the data into the Gaussian space, perform the simulation and then transform it back the original space (e.g. gold grades).

--Question:

Is Seequent using GSLIB to do geostatistics.

--Answer:

No. Seequent uses it own algorithms that has been specifically designed to run on the cloud.

--Question:

How can we integrate geophysical data into a geostatistical model

--Answer:

--Question:

What is change of support and why is it important?

--Answer:

Change of support in geostatistics refers to the process of translating spatial data or predictions from one scale to another. For instance, if you have detailed data collected at smaller spatial units (e.g., borehole samples) and you want to predict or analyze this information across larger areas or blocks (e.g., mining blocks), you need to change the support. This concept is crucial because the variability and statistical properties of the data often differ depending on the scale of measurement, and accurate modeling requires adjustments to account for these differences. Methods such as upscaling or downscaling are used to manage this transition effectively.

The change of support will change the shape and characteristics of a distribution as well as the shape of the variogram. For a linear unweighted upscaling, that is averaging, the mean of the upscaled distribution will remains the same and the variance will decrease in function of the spatial continuity as measured by the variogram.

--Question:

--Answer: